

STAT 248 Project Proposal

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The Question

Global surface temperatures have increased over the past 176 years. In this project, I study the structure of that change. Specifically, I examine whether the rate of warming has changed over time, whether the autocorrelation structure has changed (how strongly one month depends on the next), and whether residual variation contains multi-year periodicity consistent with the El Niño-Southern Oscillation (ENSO), a periodic fluctuation in sea surface temperature and atmospheric pressure across the equatorial Pacific Ocean that recurs every 2-7 years [2].

Concretely, I examine: (1) Do the autocorrelation and spectral structure differ between 1850-1949 and 1950-present? (2) After removing the long-term trend, do the residuals show statistically detectable periodic structure consistent with multi-year climate cycles?

Description of the Data

I will use the Berkeley Earth Global Average Land/Ocean Temperature Time-series from the NOAA Physical Sciences Laboratory (<https://psl.noaa.gov/data/timeseries/month/DS/BERKELEYGLBCOMT/>), originating from Berkeley Earth (<https://berkeleyearth.org/data/>).

The dataset contains:

- 2,114 monthly observations after removing unfilled future placeholders
- Monthly values: January 1850 to March 2026 (176 years)
- Temporal resolution: monthly
- Units: °C
- Outcome variable: global mean surface temperature anomaly in degrees Celsius, expressed relative to the 1951-1980 climatological baseline
- No missing values
- Single time series

The series combines Berkeley Earth's land temperature analysis with ocean temperature estimates derived from multiple sea surface temperature datasets to produce a global average. Values range from -0.804°C in the mid-19th century to $+1.481^{\circ}\text{C}$ in early 2026.

Figure 1 shows the full monthly series alongside the annual mean. The dotted vertical line at 1950 marks the boundary between the two subperiods used in the spectral analysis. The series is non-stationary, with an increasing mean over time and changes in variability across eras. Furthermore, the rate of warming also appears to accelerate after approximately 1980.

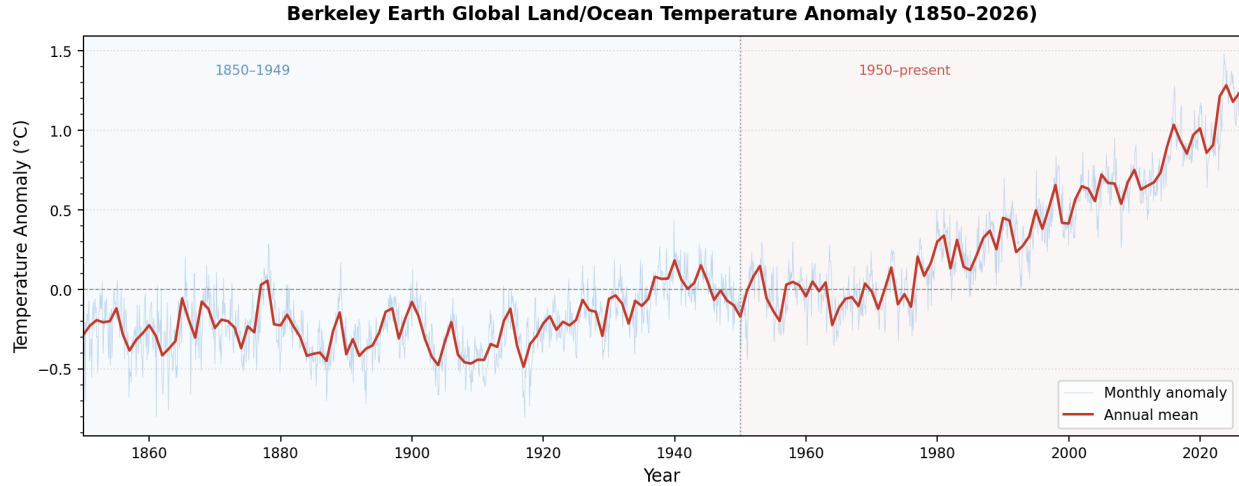


Figure 1: Berkeley Earth Global Land/Ocean Temperature Anomaly, January 1850 - March 2026. Thin blue line: monthly anomalies. Red line: annual mean. Dashed horizontal line: zero baseline (1951-1980 average). Dotted vertical line at 1950 separates the two subperiods used in the spectral analysis.

Why This is a Time Series Problem

Global temperature anomalies are correlated over time, meaning one month's temperature is related to the next. If one month is unusually warm, the next month is likely to be warm as well because the ocean and atmosphere store and release heat gradually. At longer time scales, climate patterns such as the El Niño-Southern Oscillation create multi-year fluctuations in temperature. The series also has a long-term upward trend, so the average temperature changes over time. Because of these patterns, the observations cannot be treated as independent. Therefore, ignoring the time dependence would underestimate uncertainty and hide important patterns in the data. Since this project focuses on how these patterns have changed over time, a time series framework is the natural approach.

Proposed Methods

1. **Spectral Analysis (Welch's Method).** I will compare dominant cycles in the data across two periods, from 1850-1949 and 1950-present. For each period, I will first remove the long-term trend using first differencing, then compute the power spectrum using Welch's method. This helps isolate repeating patterns rather than the long-run warming trend. I will compare periodograms to check whether key frequencies change over time, especially the annual cycle and possible multi-year signals around 2-7 years linked to El Niño-Southern Oscillation.
2. **SARIMA Modeling.** I will fit a Seasonal ARIMA model to capture both the trend and seasonal dependence in the full series. SARIMA is appropriate here because the data has a clear annual cycle and a non-stationary mean, both of which SARIMA handles through seasonal differencing and autoregressive terms. I will use ACF and PACF plots to guide model selection and choose final specifications using AIC and BIC. To evaluate predictive performance, I will assess forecast accuracy on a held-out test period (2015-2025).
3. **State Space Model with Time-Varying Trend.** I will fit a local linear trend state space model that allows both the level and slope of the trend to change over time. Unlike SARIMA, which assumes a fixed differenced structure, the state space model can estimate whether the rate of warming has accelerated or slowed across the record. I will compare its forecast accuracy against the SARIMA model on the same test period to assess whether allowing a time-varying trend improves prediction.

Expected Outcomes

For a positive result, I expect both periods to show a strong annual cycle in the spectral analysis. I also expect the modern period to show more low-frequency variation, potentially consistent with multi-year climate patterns such as the El Niño-Southern Oscillation, along with stronger and more persistent autocorrelation than in the early period. For the state space model, I expect the estimated trend to show changes in the rate of warming over time and to outperform SARIMA in forecasting if the trend is not constant.

A null result would show similar spectral patterns across both periods after detrending, with no clear increase in low-frequency power. The autocorrelation structure would also be similar across eras, and the state space model would not improve forecasting performance compared to SARIMA, suggesting that a simpler stationary structure is sufficient after accounting for the trend.

I believe either outcome is meaningful. The results would clarify whether global temperature behaves like a stable time series with a changing mean, or whether its dependence structure has also evolved over time.

References

- [1] Rohde, R. A. and Hausfather, Z.: The Berkeley Earth Land/Ocean Temperature Record, *Earth System Science Data*, 2020.
- [2] National Centers for Environmental Information. *ENSO: Recent Evolution, Current Status and Predictions*. NOAA. <https://www.ncei.noaa.gov/access/monitoring/enso/>